

ibeam

July 25, 2025

```
[63]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.metrics import mean_absolute_error, mean_squared_error, \
    mean_absolute_error, r2_score
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.model_selection import GridSearchCV, RandomizedSearchCV

plt.rcParams["font.family"] = "Times New Roman"
plt.rcParams['text.usetex'] = True
```

```
[64]: df = pd.read_csv('Ratio.csv')
df.head()
```

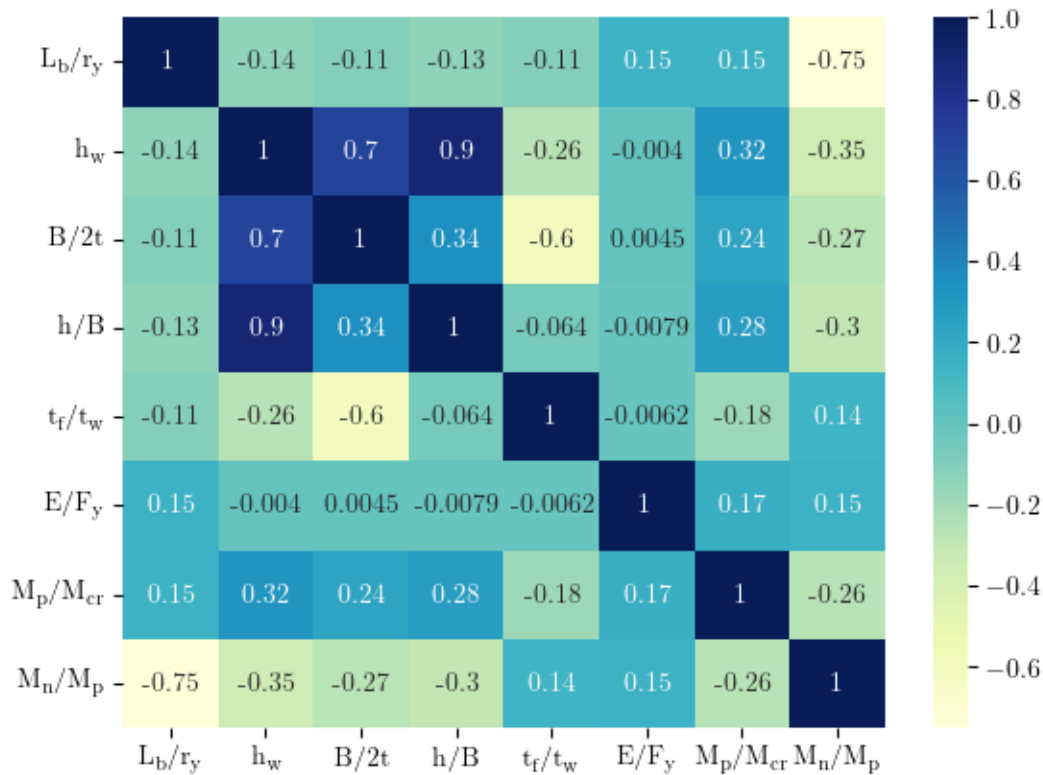
```
[64]:
```

	Lb/ry	h/tw	B/2t	h/B	tf/tw	E/Fy	Mp/Mcr	\
0	45.845272	41.0	4.491525	2.655975	1.718447	208.279468	0.558946	
1	54.441261	41.0	4.491525	2.655975	1.718447	208.279468	0.789461	
2	65.902579	41.0	4.491525	2.655975	1.718447	208.279468	0.978898	
3	74.498567	41.0	4.491525	2.655975	1.718447	208.279468	1.248814	
4	85.959885	41.0	4.491525	2.655975	1.718447	208.279468	1.607064	

	Mn/Mp
0	0.843663
1	0.760744
2	0.647926
3	0.570788
4	0.483207

```
[65]: tempdf = df.rename(columns=
    {"Lb/ry": r"L\textsubscript{b}/r\textsubscript{y}",
     "h/tw" : r"h\textsubscript{w}",
     "tf/tw" : r"t\textsubscript{f}/t\textsubscript{w}",
     "E/Fy" : r"E/F\textsubscript{y}",
     "Mp/Mcr" : r"M\textsubscript{p}/M\textsubscript{cr}",
     "Mn/Mp" : r"M\textsubscript{n}/M\textsubscript{p}"
    }, errors="raise")
```

```
sns.heatmap(tempdf.corr(numeric_only=True), cmap="YlGnBu", annot=True)
plt.savefig(f'crosscorrelation.png', dpi=640, bbox_inches='tight')
```



```
[66]: df_stats = df.describe().transpose()
df_stats = df_stats[['mean', 'std']]
df_stats['variance'] = df_stats['std'] ** 2
df_stats
```

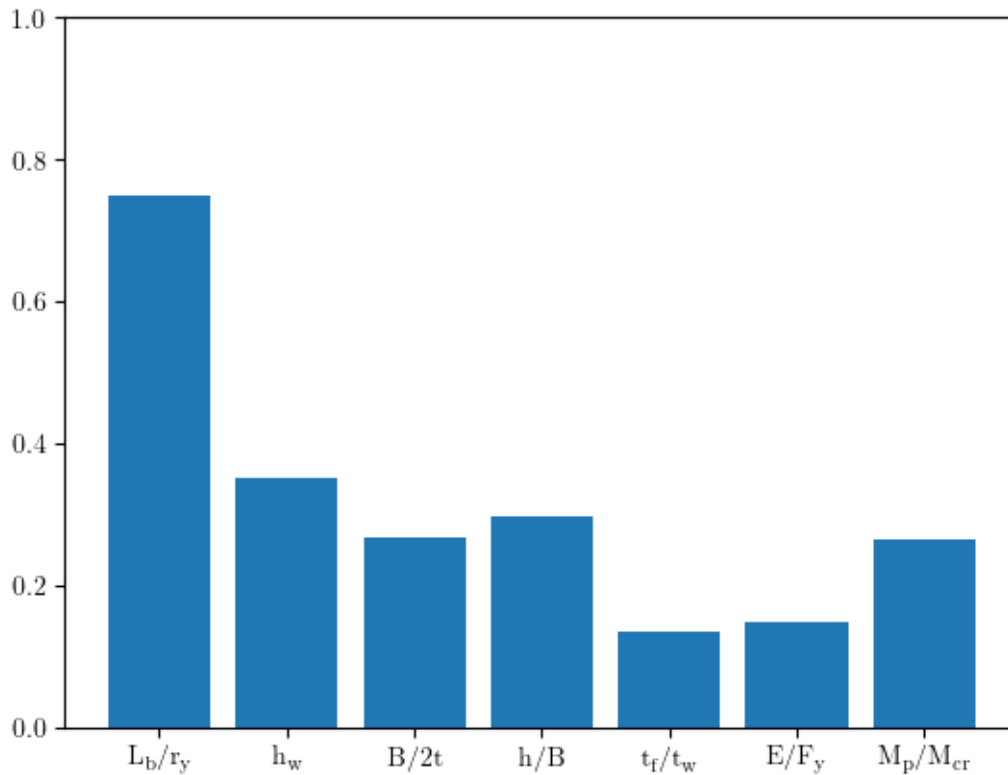
```
[66]:
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	mean	std	variance
Lb/ry	97.366596	37.083167	1375.161280
h/tw	34.963695	14.070018	197.965420
B/2t	5.634730	1.363515	1.859172
h/B	1.853852	0.583968	0.341018
tf/tw	1.653545	0.116217	0.013506
E/Fy	248.937214	40.796326	1664.340176
Mp/Mcr	1.624280	0.948242	0.899163
Mn/Mp	0.564298	0.189916	0.036068

```
[68]: C = tempdf.corr(numeric_only=True)[r'M\textsubscript{n}/M\textsubscript{p}'][:
↪-1]
plt.bar(C.index, np.abs(C.values))
```

```
plt.ylim([0, 1])

plt.savefig(f'correlation-with-Mn-Mp.png', dpi=640, bbox_inches='tight')
```



```
[69]: X_full = df[['Lb/ry', 'h/tw', 'B/2t', 'h/B', 'tf/tw', 'E/Fy', 'Mp/Mcr']]
      y_full = df['Mn/Mp']
```

```
[70]: from sklearn.model_selection import train_test_split

      X_train, X_test, y_train, y_test = train_test_split(X_full, y_full, test_size=0.
      ↪2, random_state=123)
```

```
[71]: def evaluate_pipeline(pipeline, threshold=0.2):
      pipeline_train_prd = pipeline.predict(X_train)
      pipeline_train_outliers = np.abs(pipeline_train_prd - y_train) > threshold
      mod_pipeline_train_prd = pipeline_train_prd[~ pipeline_train_outliers]
      mod_y_train = y_train[~ pipeline_train_outliers]
      pipeline_train_mse = mean_squared_error(mod_pipeline_train_prd, mod_y_train)
      pipeline_train_mae = mean_absolute_error(mod_pipeline_train_prd,
      ↪mod_y_train)
      pipeline_train_r2_score = r2_score(mod_pipeline_train_prd, mod_y_train)
```

```

pipeline_test_prd = pipeline.predict(X_test)
pipeline_test_outliers = np.abs(pipeline_test_prd - y_test) > threshold
mod_pipeline_test_prd = pipeline_test_prd[~ pipeline_test_outliers]
mod_y_test = y_test[~ pipeline_test_outliers]
pipeline_test_mse = mean_squared_error(mod_pipeline_test_prd, mod_y_test)
pipeline_test_mae = mean_absolute_error(mod_pipeline_test_prd, mod_y_test)
pipeline_test_r2_score = r2_score(mod_pipeline_test_prd, mod_y_test)

pipeline_full_prd = pipeline.predict(X_full)
pipeline_full_outliers = np.abs(pipeline_full_prd - y_full) > threshold
mod_pipeline_full_prd = pipeline_full_prd[~ pipeline_full_outliers]
mod_y_full = y_full[~ pipeline_full_outliers]
pipeline_full_mse = mean_squared_error(mod_pipeline_full_prd, mod_y_full)
pipeline_full_mae = mean_absolute_error(mod_pipeline_full_prd, mod_y_full)
pipeline_full_r2_score = r2_score(mod_pipeline_full_prd, mod_y_full)

metrics = {
    'y_train': mod_y_train,
    'y_test' : mod_y_test,
    'train_mse': pipeline_train_mse,
    'train_mae': pipeline_train_mae,
    'train_r2': pipeline_train_r2_score,
    'test_mse': pipeline_test_mse,
    'test_mae': pipeline_test_mae,
    'test_r2': pipeline_test_r2_score,
    'full_mse': pipeline_full_mse,
    'full_mae': pipeline_full_mae,
    'full_r2': pipeline_full_r2_score,
    'train_predictions': mod_pipeline_train_prd,
    'test_predictions': mod_pipeline_test_prd
}

return metrics

def plot_accuracy(pipeline_metrics, title):
    print(title)
    print(pipeline_metrics)
    y_train = pipeline_metrics['y_train']
    y_test = pipeline_metrics['y_test']

    pipeline_train_prd = pipeline_metrics['train_predictions']
    pipeline_test_prd = pipeline_metrics['test_predictions']
    pipeline_train_r2_score = pipeline_metrics['train_r2']
    pipeline_test_r2_score = pipeline_metrics['test_r2']
    pipeline_full_r2_score = pipeline_metrics['full_r2']

```

```

fig_train, axes = plt.subplots(figsize=(4, 4))
axes.plot([0, 1], [0, 1])
axes.text(0.1, 0.7, f'$R^2={pipeline_train_r2_score:.5f}$')
axes.plot(y_train, pipeline_train_prd, 'bo', alpha=0.5, markeredgewidth=0.
↳1, label='Train data');
axes.set_xlabel('Actual Value')
axes.set_ylabel('Predicted Value')
axes.legend()
fig_train.savefig(f'{title}/{title}-train.png', dpi=640,
↳bbox_inches='tight')

fig_test, axes = plt.subplots(figsize=(4, 4))
axes.plot([0, 1], [0, 1])
axes.text(0.1, 0.7, f'$R^2={pipeline_test_r2_score:.5f}$')
axes.plot(y_test, pipeline_test_prd, 'ro', alpha=0.5, markeredgewidth=0.1,
↳label='Test data');
axes.set_xlabel('Actual Value')
axes.set_ylabel('Predicted Value')
axes.legend()
fig_test.savefig(f'{title}/{title}-test.png', dpi=640, bbox_inches='tight')

fig_full, axes = plt.subplots(figsize=(4, 4))
axes.plot([0, 1], [0, 1])
axes.text(0.1, 0.7, f'$R^2={pipeline_full_r2_score:.5f}$')
axes.plot(y_train, pipeline_train_prd, 'bo', alpha=0.5, markeredgewidth=0.
↳1, label='Train data');
axes.plot(y_test, pipeline_test_prd, 'ro', alpha=0.5, markeredgewidth=0.1,
↳label='Test data');
axes.set_xlabel('Actual Value')
axes.set_ylabel('Predicted Value')
axes.legend()
fig_full.savefig(f'{title}/{title}-full.png', dpi=640, bbox_inches='tight')

def plot_cross_validation(cv_result, titlename):
    mean_mse = cv_result['mean_test_score']
    criterions = cv_result['params'][0].keys()

    params_df = pd.DataFrame(cv_result['params'].tolist())
    heatmap_data = params_df.copy()
    heatmap_data['mean_r2'] = mean_mse
    if all(mean_mse < 0):
        heatmap_data['mean_r2'] = -mean_mse

    for col in params_df.columns:
        df = heatmap_data.groupby(col)['mean_r2'].mean()
        metrics = df.keys().tolist()

```

```

values = df.values.tolist()

angles = np.linspace(0, 2 * np.pi, len(metrics), endpoint=False).
↳tolist()

fig, ax = plt.subplots(figsize=(6, 6), subplot_kw=dict(polar=True))
# print(np.max(values) * 1.1)
# ax.set_rlim(np.max(values) * 1.1)
ax.margins(0.1)
ax.scatter(angles, values)
ax.fill(angles, values, alpha=0.25)

# Set axis labels
ax.set_xticks(angles)
ax.set_xticklabels(metrics)

plt.savefig(f'{titlename}/{titlename}-hyperparam-{col}.png', dpi=640,
↳bbox_inches='tight')

```

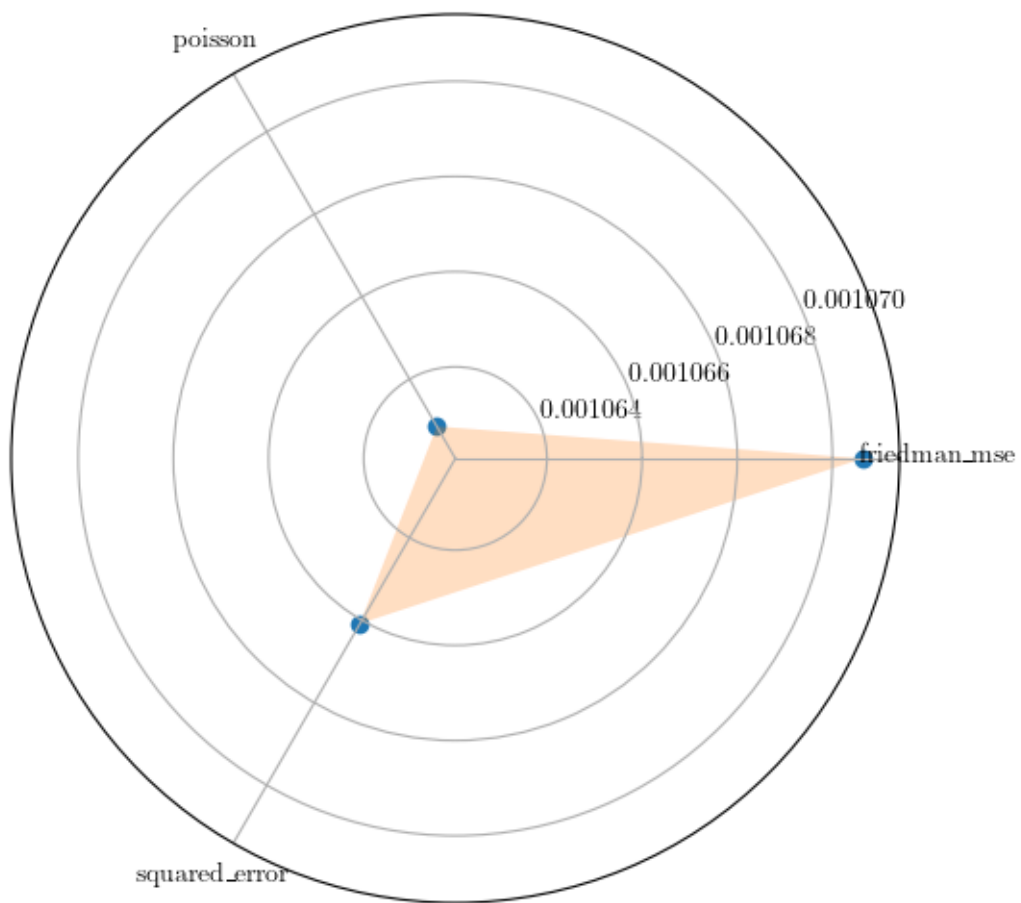
```

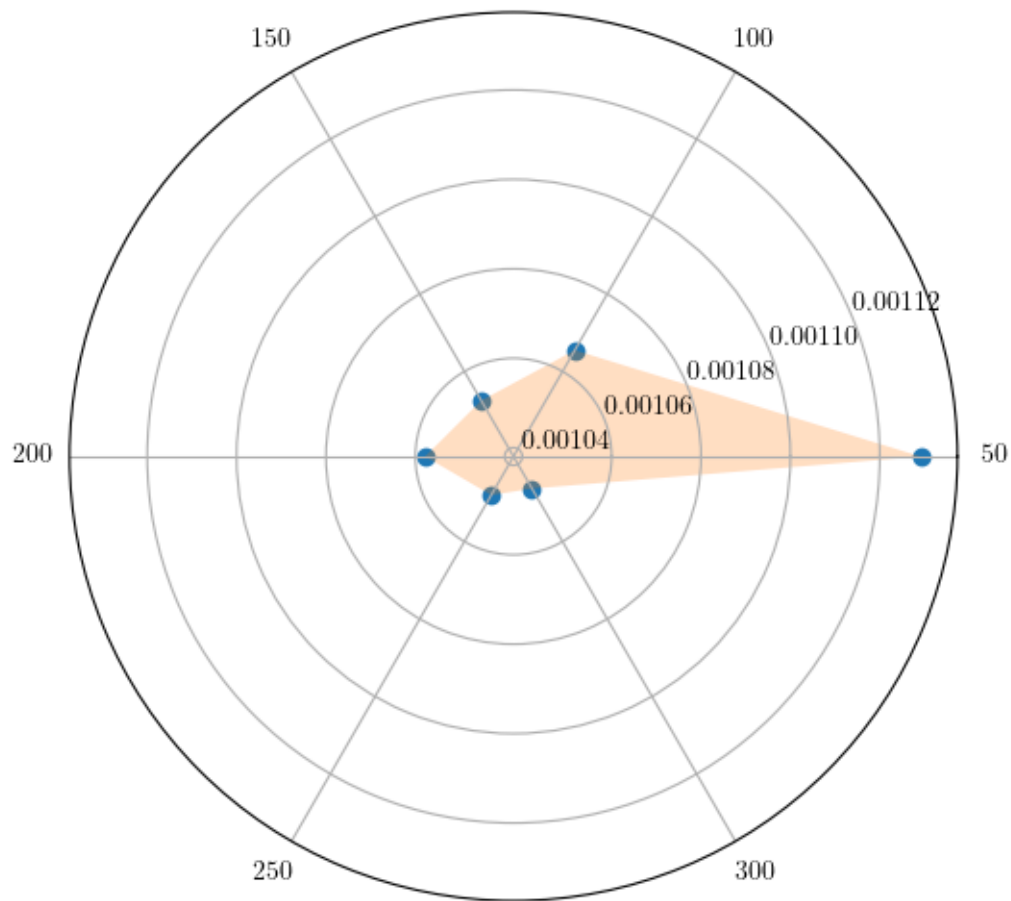
[72]: rf_params = {
        'criterion' : ['squared_error', 'friedman_mse', 'poisson'],
        'n_estimators' : [50, 100, 150, 200, 250, 300],
    }

rf_cv = GridSearchCV(RandomForestRegressor(random_state=42), rf_params,
↳scoring='neg_mean_squared_error')
rf_cv.fit(X_train, y_train)
rf_best = rf_cv.best_estimator_
rf_metrics = evaluate_pipeline(rf_best)

rf_cv_result = pd.DataFrame(rf_cv.cv_results_)
plot_cross_validation(rf_cv_result, 'random-forest')

```





```
[73]: rf_metrics = evaluate_pipeline(rf_best)
      plot_accuracy(rf_metrics, 'random-forest')
```

```
random-forest
{'y_train': 21      0.587692
430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
Name: Mn/Mp, Length: 352, dtype: float64, 'y_test': 13      0.301036
125      0.643799
287      0.409213
204      0.552481
```

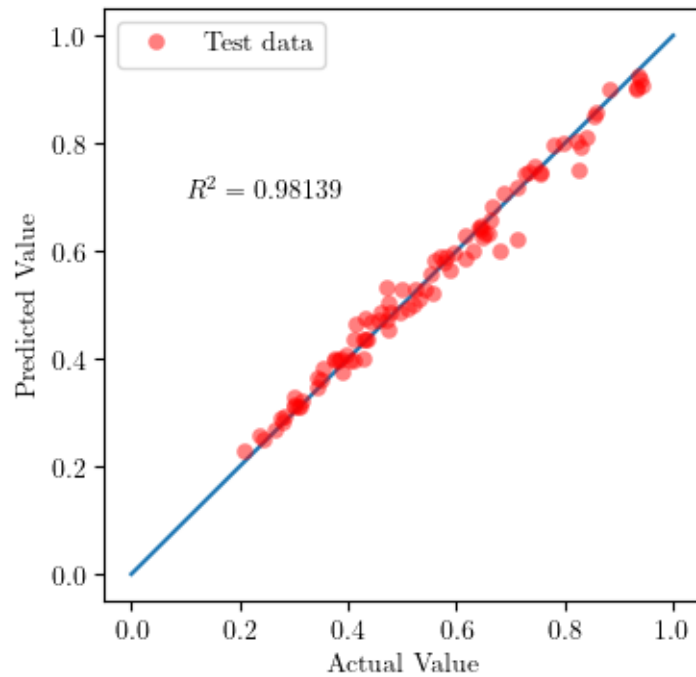
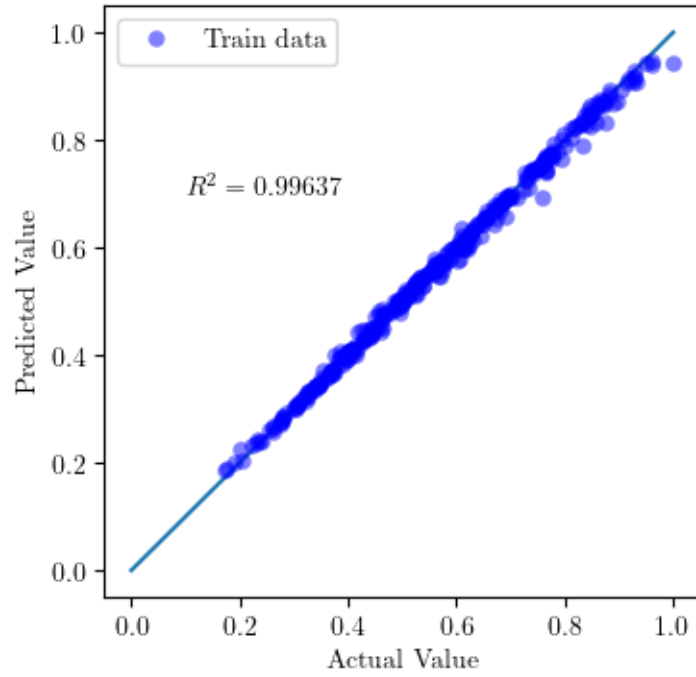
172 0.615535
...
199 0.352128
31 0.386317
240 0.753010
59 0.382423
328 0.426775

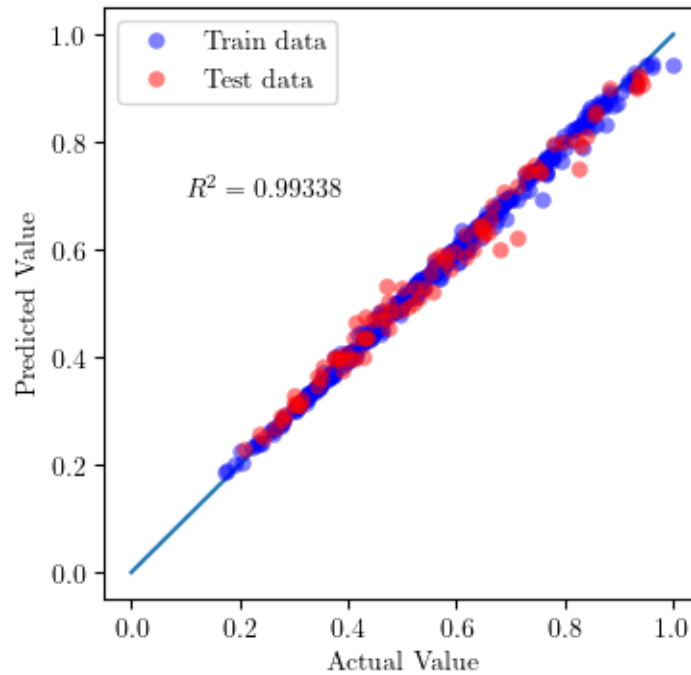
Name: Mn/Mp, Length: 89, dtype: float64, 'train_mse': 0.00012393592713143433,
'train_mae': 0.007611368377886381, 'train_r2': 0.9963729579588495, 'test_mse':
0.0006322251720869567, 'test_mae': 0.01873371248382027, 'test_r2':
0.9813861508234758, 'full_mse': 0.00022651584278005453, 'full_mae':
0.009856013787020433, 'full_r2': 0.9933752199217437, 'train_predictions':

array([0.58021218, 0.27146172, 0.6309638 , 0.69739713, 0.79005876,
0.54929841, 0.41017075, 0.64484248, 0.4019365 , 0.33092088,
0.74815266, 0.34663822, 0.90723068, 0.77100886, 0.59936387,
0.88751296, 0.34429944, 0.54563486, 0.32717906, 0.84242038,
0.39866352, 0.26912558, 0.74105836, 0.85780467, 0.44289106,
0.40203257, 0.35824119, 0.77685321, 0.91298318, 0.59572313,
0.41896979, 0.47391966, 0.94817178, 0.68561016, 0.57191265,
0.3435953 , 0.68039247, 0.52908735, 0.43654636, 0.53813045,
0.81257862, 0.3462899 , 0.33598026, 0.51398966, 0.48209385,
0.67010171, 0.87523402, 0.61436354, 0.5960688 , 0.66926265,
0.57011424, 0.65415491, 0.49285289, 0.83731377, 0.59978039,
0.74584318, 0.53034426, 0.51934848, 0.37104149, 0.45096627,
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0.58598112, 0.23141341, 0.4811219 , 0.51540093, 0.51235323,
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 0.86641333, 0.69220306, 0.55889986, 0.34345716, 0.77722551,
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 0.5864979 , 0.56402895]), 'test_predictions': array([0.33035874,
 0.64572979, 0.39611527, 0.55868336, 0.58598787,
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 0.34599596, 0.22873765, 0.46628571, 0.64413296, 0.57835736,
 0.60003765, 0.26892829, 0.36302543, 0.90346538, 0.5334159 ,
 0.48708379, 0.85650322, 0.37505111, 0.31276208, 0.31030905,
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 0.80148781, 0.85108822, 0.60086284, 0.43630311, 0.28338941,
 0.40603902, 0.4870655 , 0.5904735 , 0.39516853, 0.8036984 ,
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 0.43611081, 0.43621424, 0.90588533, 0.51251555, 0.63153842,
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0.92713266, 0.32095995, 0.63365556, 0.52830433, 0.50126538,
0.31640125, 0.52750843, 0.68384859, 0.59644311, 0.7426069 ,
0.47076926, 0.89907056, 0.29366858, 0.43544769, 0.5834293 ,
0.36090419, 0.79844021, 0.81035224, 0.53032641, 0.38089783,
0.40009528, 0.7442075 , 0.3967314 , 0.40019964]})}





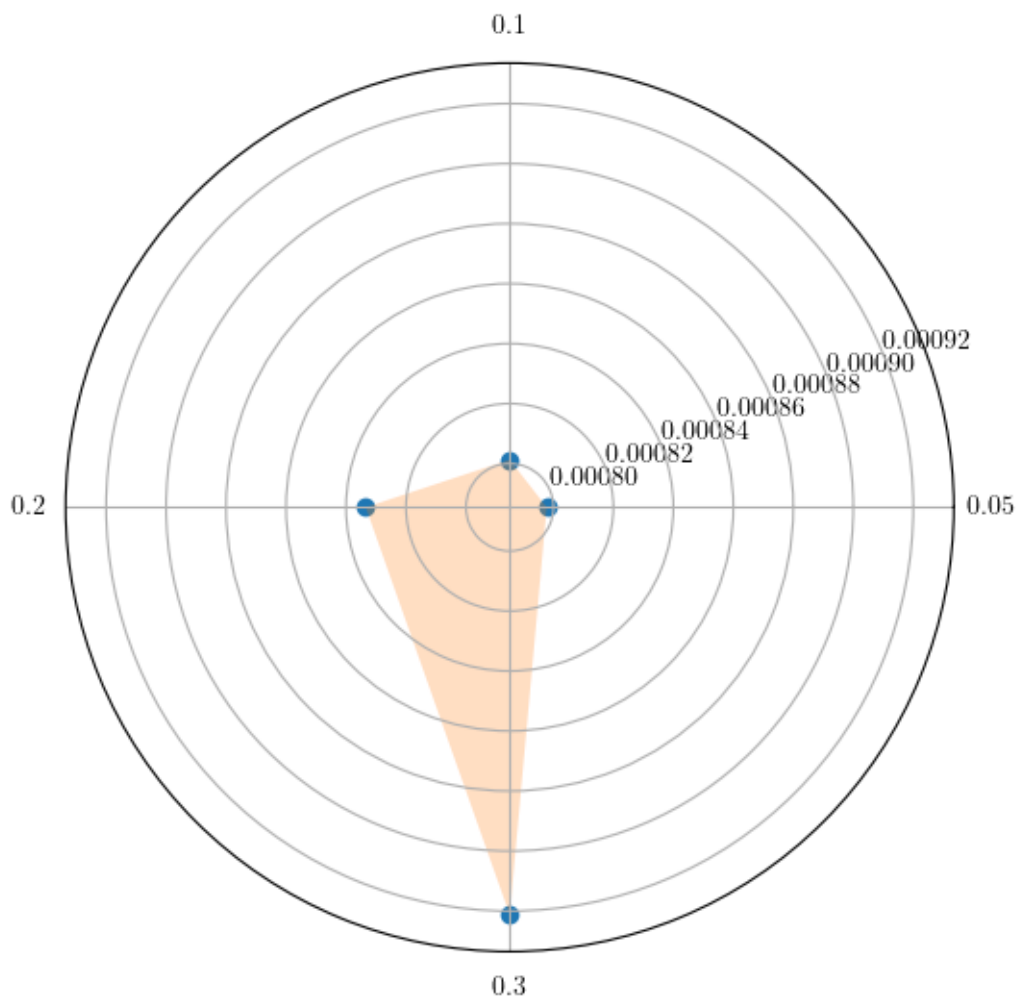
```
[74]: gb_params = {
    'subsample' : [0.7, 0.8, 0.9, 1.0],
    'n_estimators' : [150, 200, 250, 300, 350],
    'learning_rate' : [0.05, 0.1, 0.2, 0.3],
    'max_depth' : [2, 4, 8, 16, 32],
}

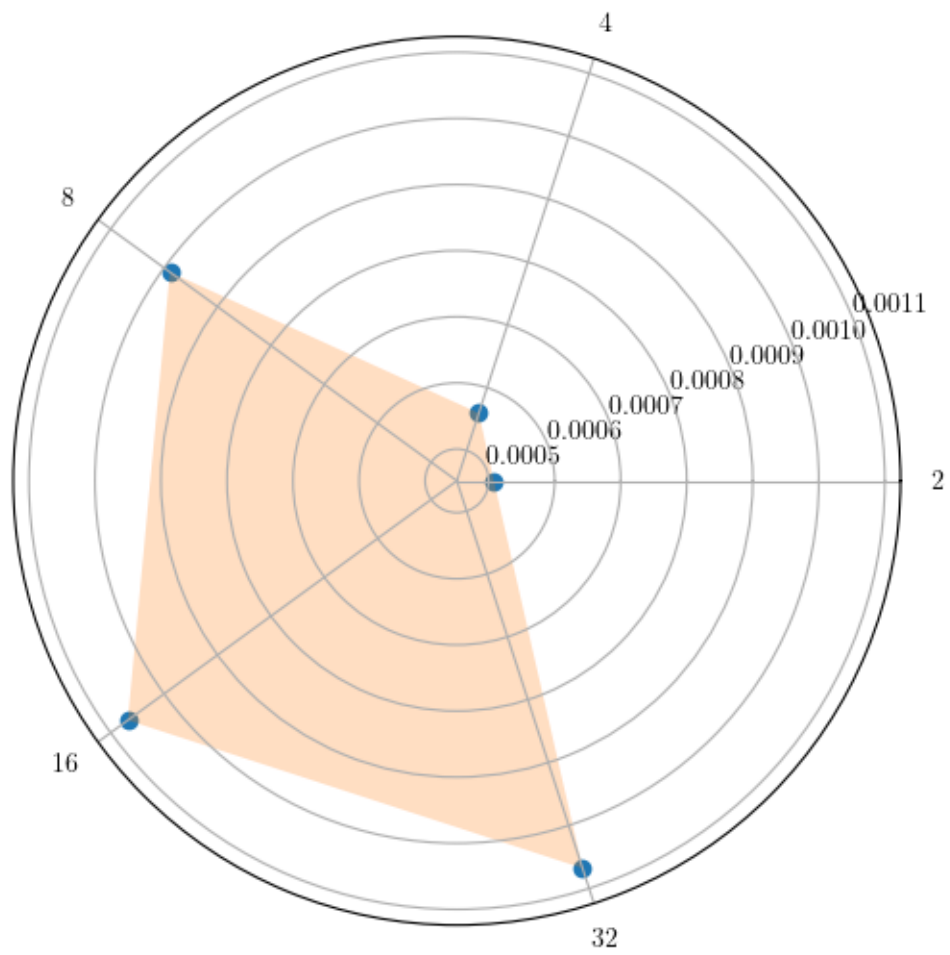
gb_cv = GridSearchCV(GradientBoostingRegressor(random_state=42), gb_params,
    ↪scoring='neg_mean_squared_error')
gb_cv.fit(X_train, y_train)
gb_best = gb_cv.best_estimator_

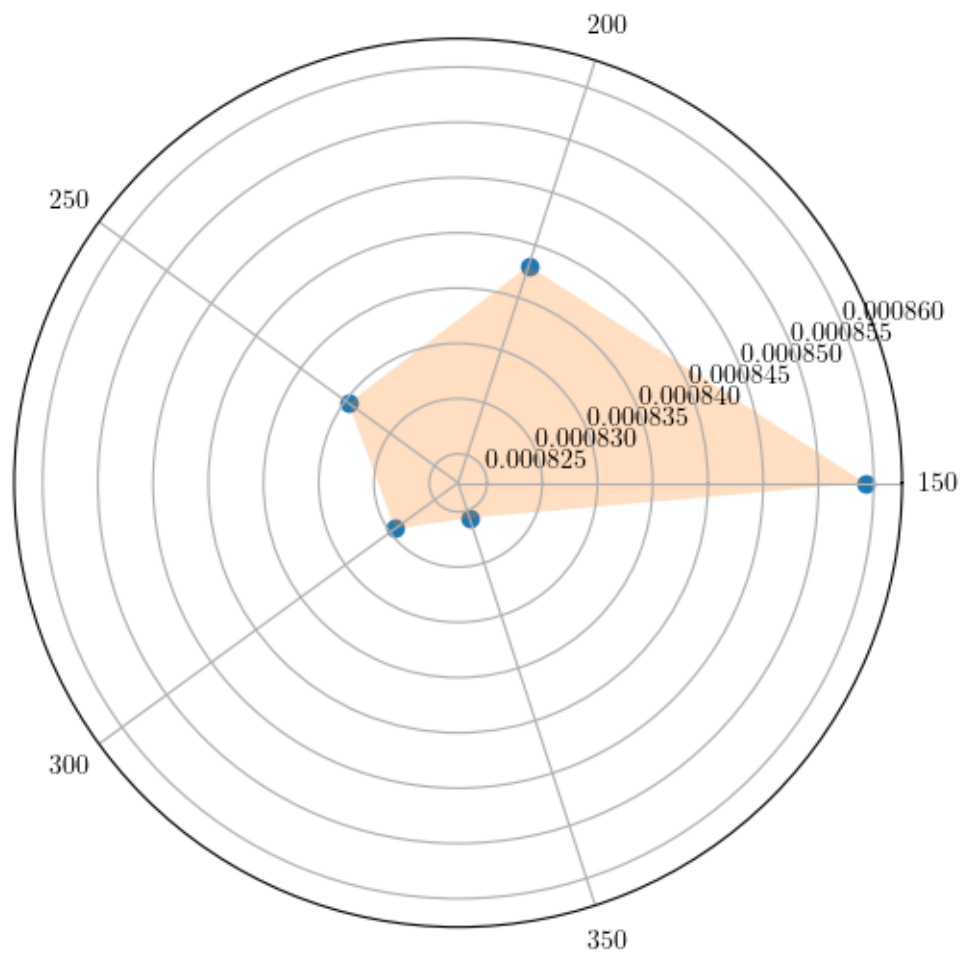
print(gb_cv.best_params_)

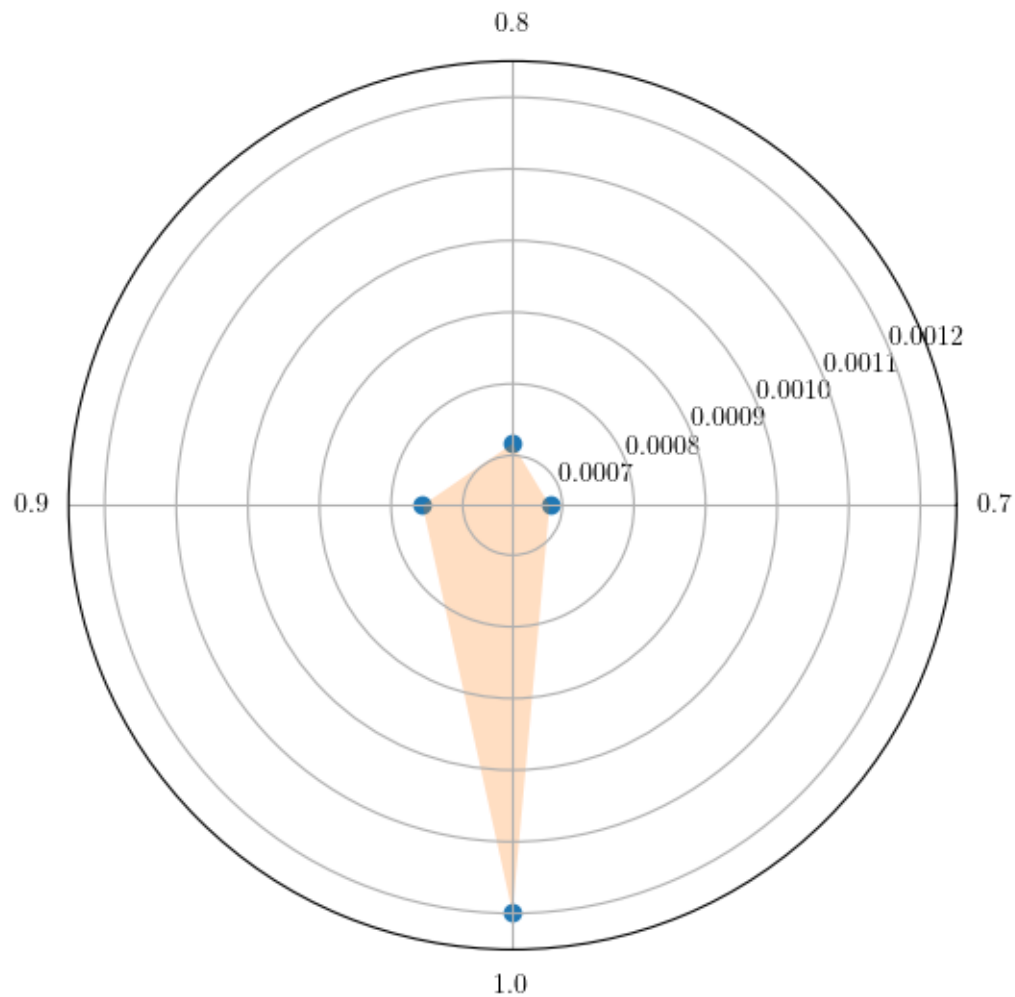
gb_cv_result = pd.DataFrame(gb_cv.cv_results_)
plot_cross_validation(gb_cv_result, 'gradient-boosting')

{'learning_rate': 0.1, 'max_depth': 2, 'n_estimators': 350, 'subsample': 0.7}
```









```
[75]: gb_metrics = evaluate_pipeline(gb_best, 100)
      plot_accuracy(gb_metrics, 'gradient-boosting')
```

```
gradient-boosting
{'y_train': 21      0.587692
430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
Name: Mn/Mp, Length: 352, dtype: float64, 'y_test': 13      0.301036
```

```

125    0.643799
287    0.409213
204    0.552481
172    0.615535

...
199    0.352128
31     0.386317
240    0.753010
59     0.382423
328    0.426775

```

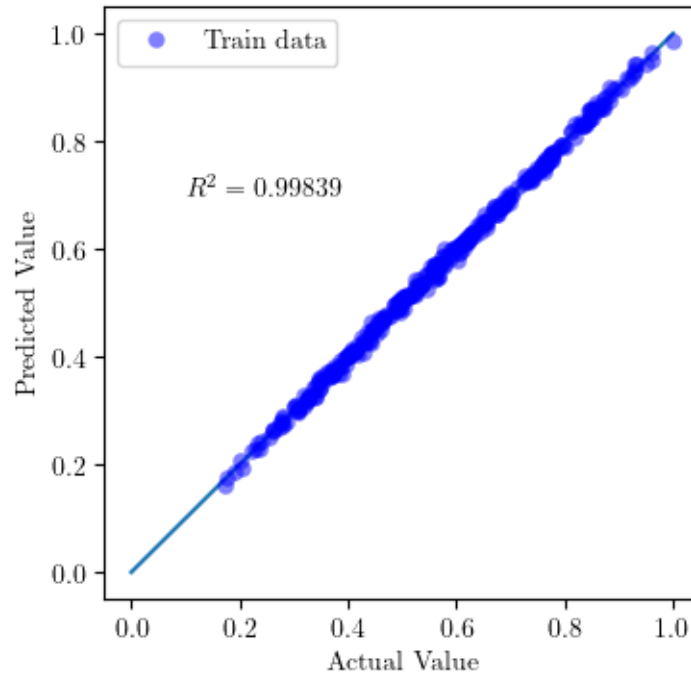
```

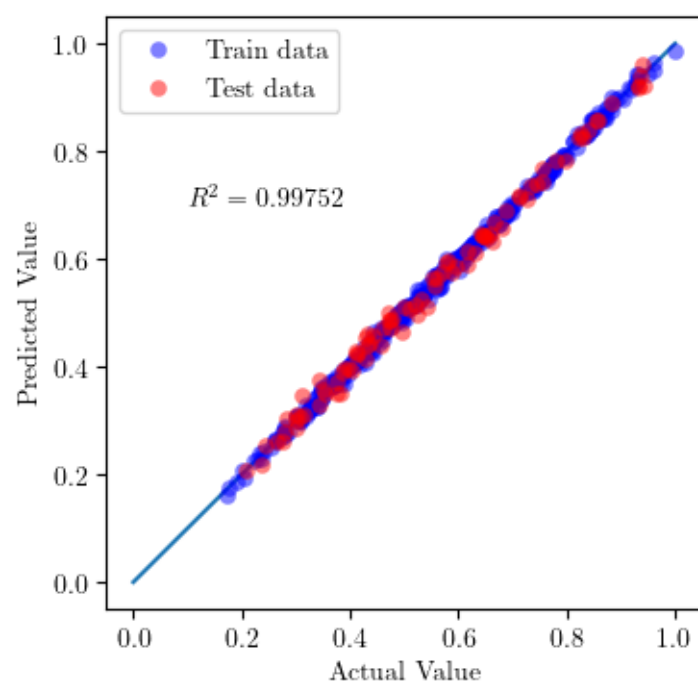
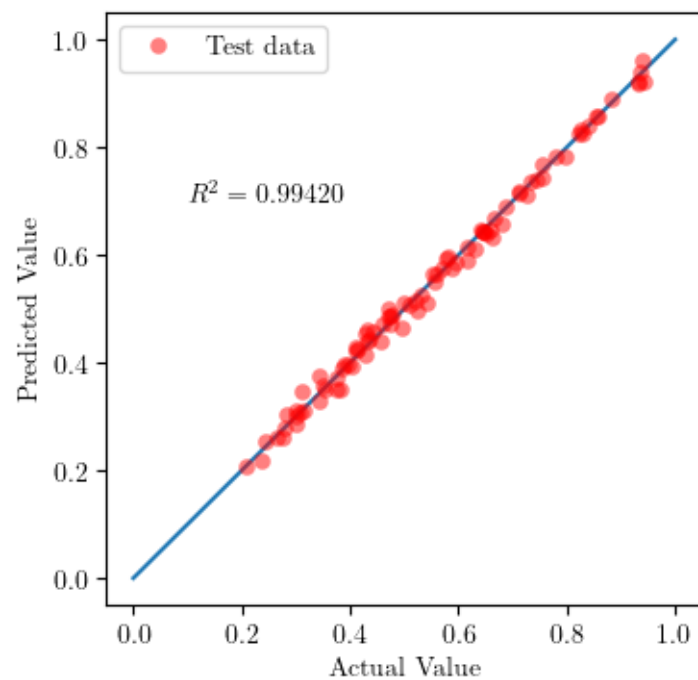
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0.39012053, 0.76659786, 0.34923423, 0.4150639]})





```
[76]: from sklearn.svm import LinearSVR, SVR
from scipy.stats import loguniform, uniform
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

kernels = ['linear', 'rbf', 'poly']
svr_params = {
    'svr__C': loguniform(1e-1, 1e2),
    'svr__epsilon': uniform(0.01, 0.1)
}

svr_metrics = dict()

for kernel in ['linear', 'rbf', 'poly']:
    if kernel == 'linear':
        base_model = SVR(kernel=kernel)
        base_params = svr_params
    else:
        base_model = SVR(kernel=kernel)
        base_params = {**svr_params, 'svr__gamma': loguniform(1e-3, 1e3)}

    svr_pipeline = Pipeline([
        ('scaler', StandardScaler()),
        ('svr', base_model)
    ])

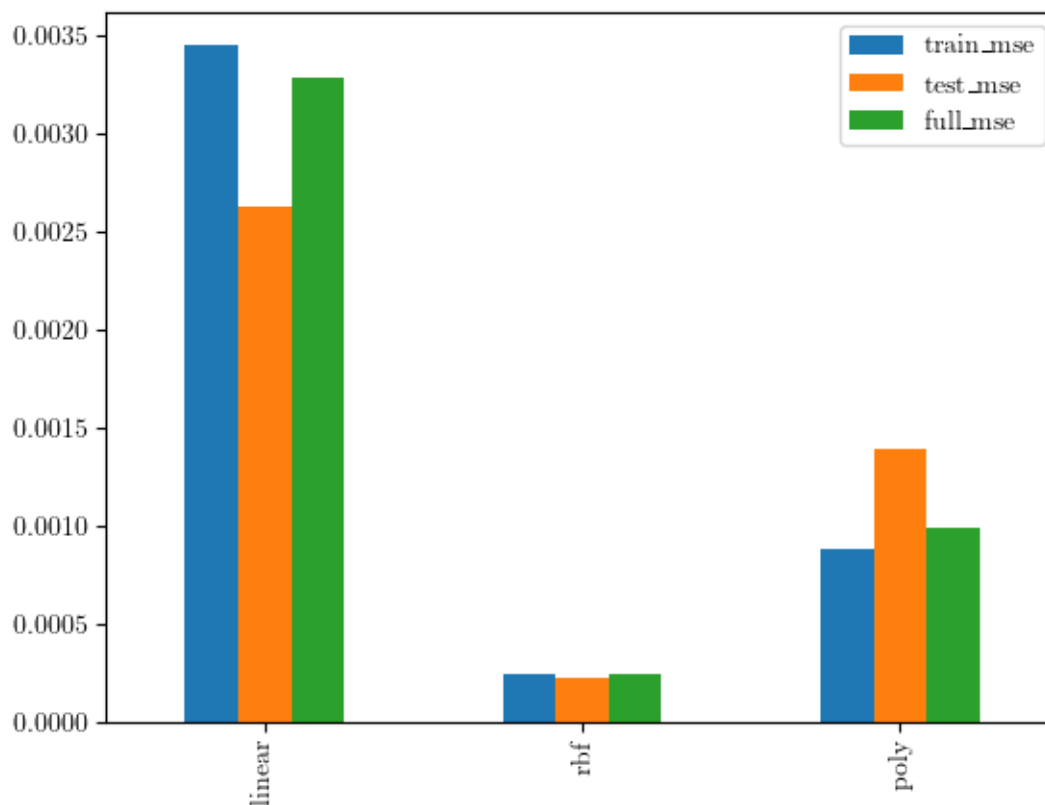
    svr_cv = RandomizedSearchCV(svr_pipeline, base_params, n_iter=10,
    random_state=100, n_jobs=-1, scoring='neg_mean_squared_error')
    svr_cv.fit(X_train, y_train)

    svr_best = svr_cv.best_estimator_
    svr_cv_result = pd.DataFrame(svr_cv.cv_results_)
    print(svr_cv.best_params_)

    svr_metrics[kernel] = evaluate_pipeline(svr_best)

{'svr__C': np.float64(0.2571104740261944), 'svr__epsilon':
np.float64(0.067509332942725)}
{'svr__C': np.float64(34.2238132147075), 'svr__epsilon':
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np.float64(0.037836938509379615), 'svr__gamma': np.float64(0.3524565164435525)}

[77]: pd.DataFrame(svr_metrics).filter(regex='_mse', axis=0).transpose().plot.bar()
plt.savefig('svr/svr-kernel-mse.png', dpi=640, bbox_inches='tight')
```



```
[78]: for kernel in kernels:
      plot_accuracy(svr_metrics[kernel], f'SVR-{kernel}')
```

```
SVR-linear
{'y_train': 21      0.587692
430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
Name: Mn/Mp, Length: 348, dtype: float64, 'y_test': 13      0.301036
125      0.643799
287      0.409213
204      0.552481
172      0.615535
...
```

```

199    0.352128
31     0.386317
240    0.753010
59     0.382423
328    0.426775
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```

```

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SVR-rbf
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430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
Name: Mn/Mp, Length: 352, dtype: float64, 'y_test': 13      0.301036
125      0.643799
287      0.409213
204      0.552481
172      0.615535
...
199      0.352128
31       0.386317
240      0.753010
59       0.382423
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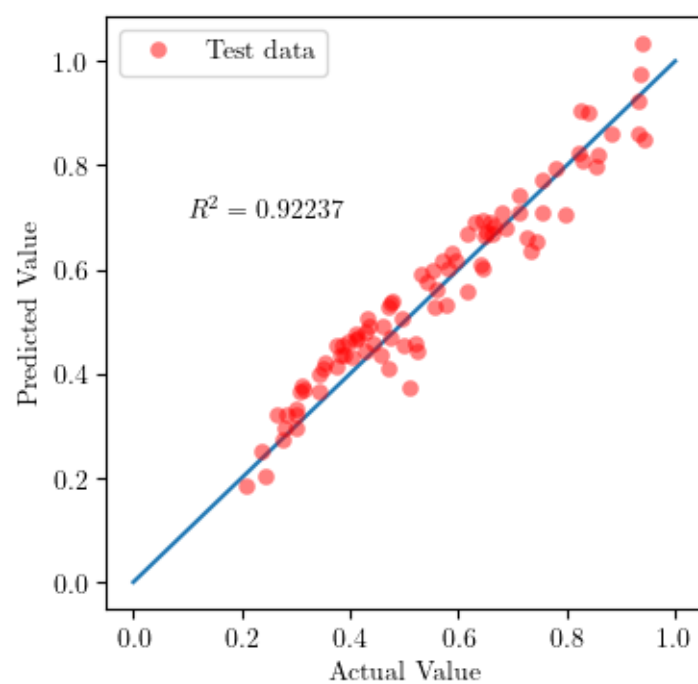
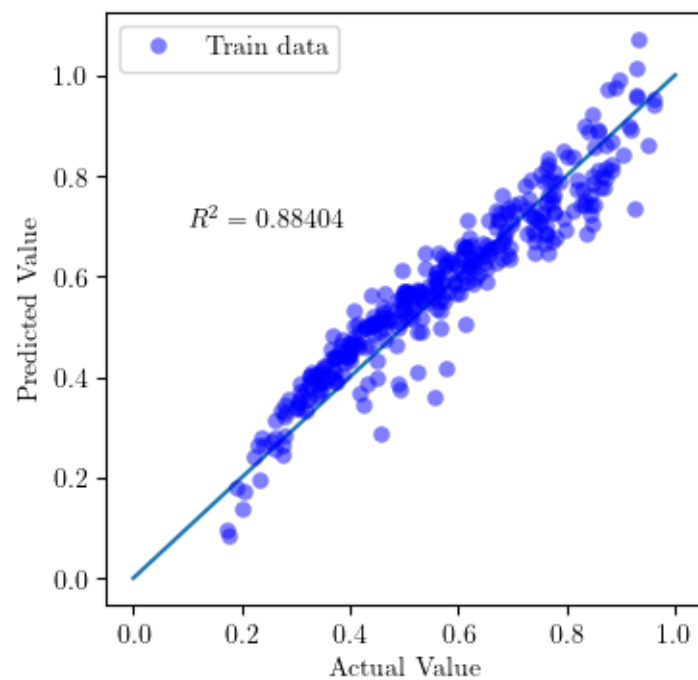
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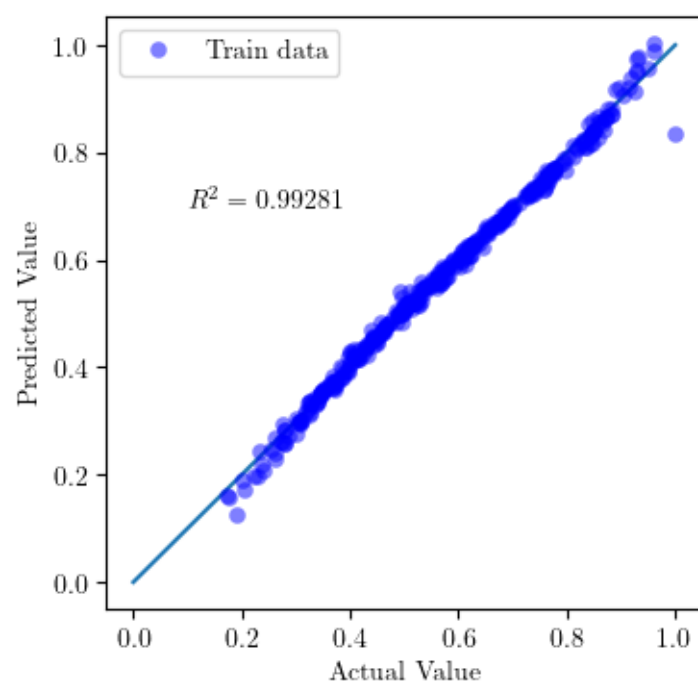
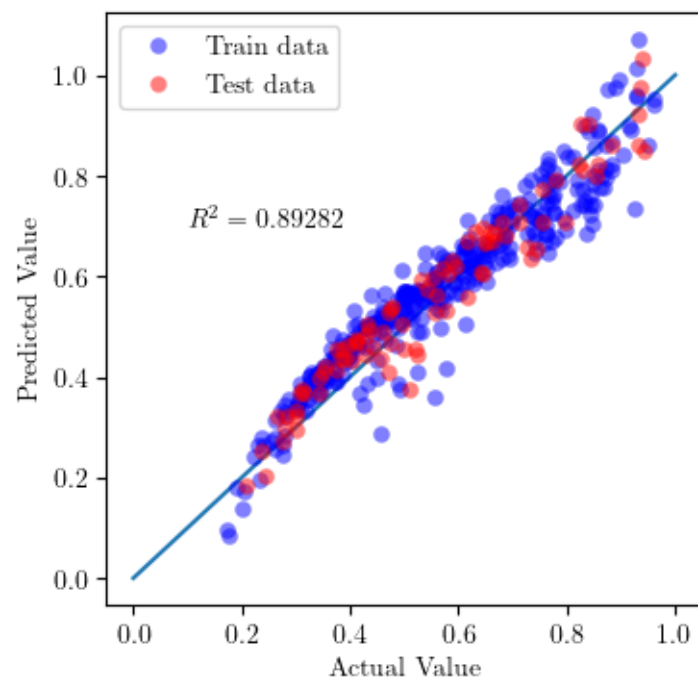
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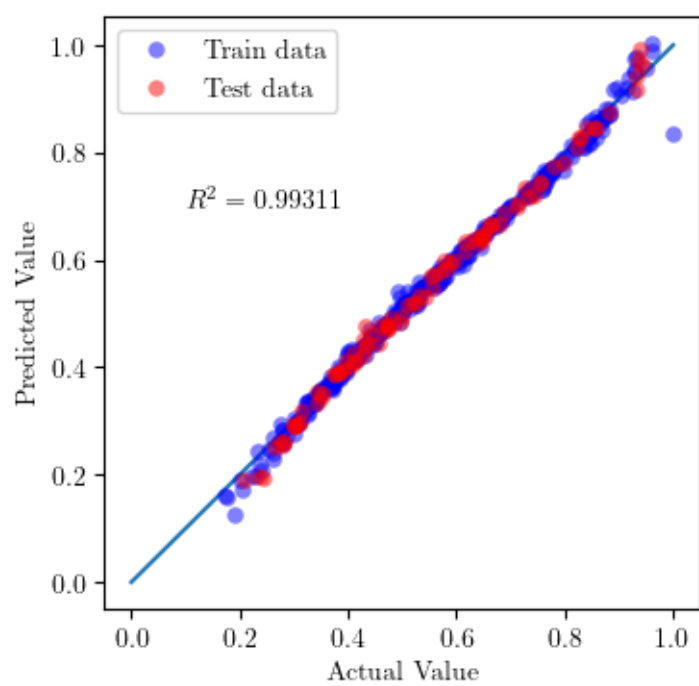
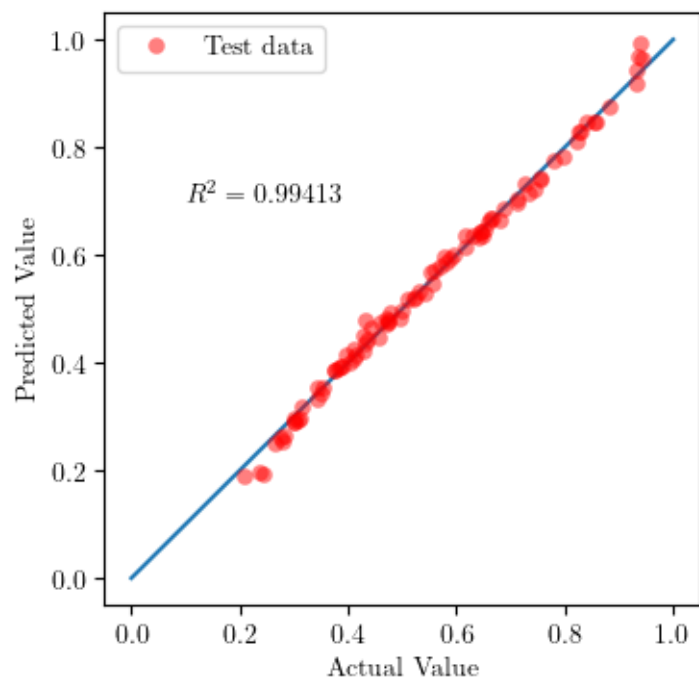
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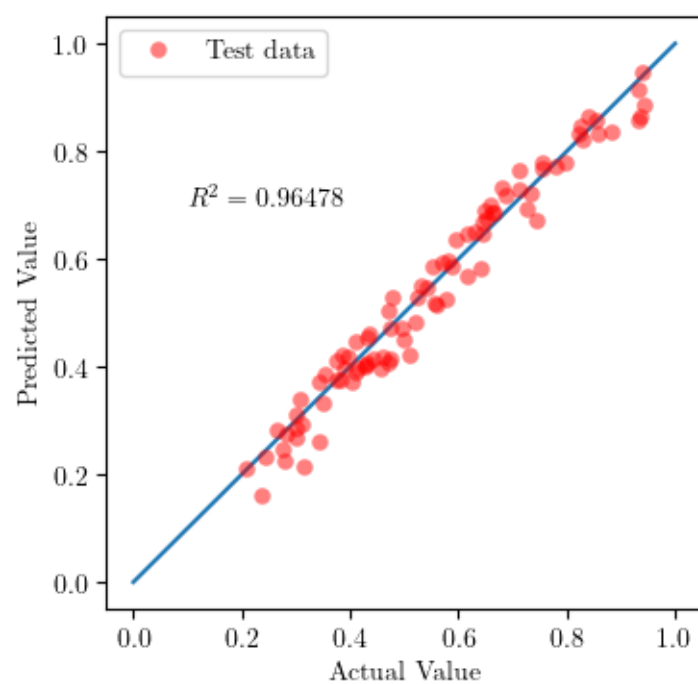
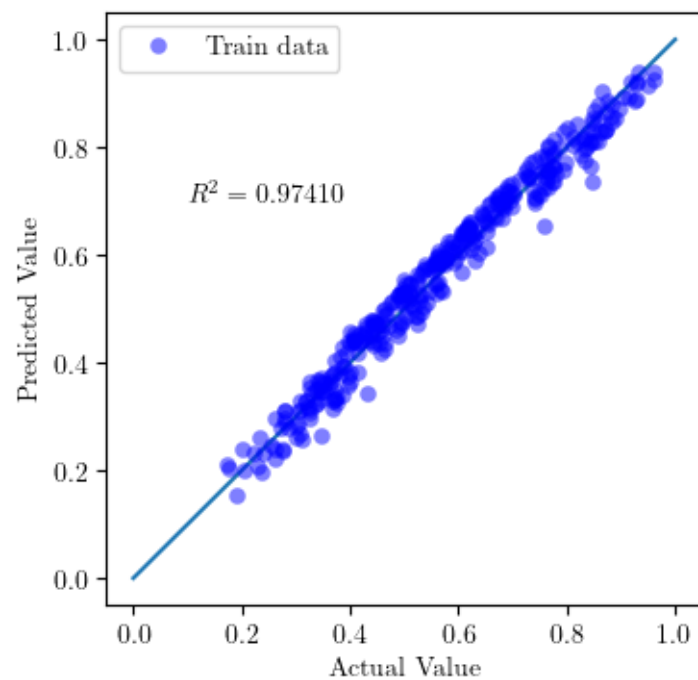
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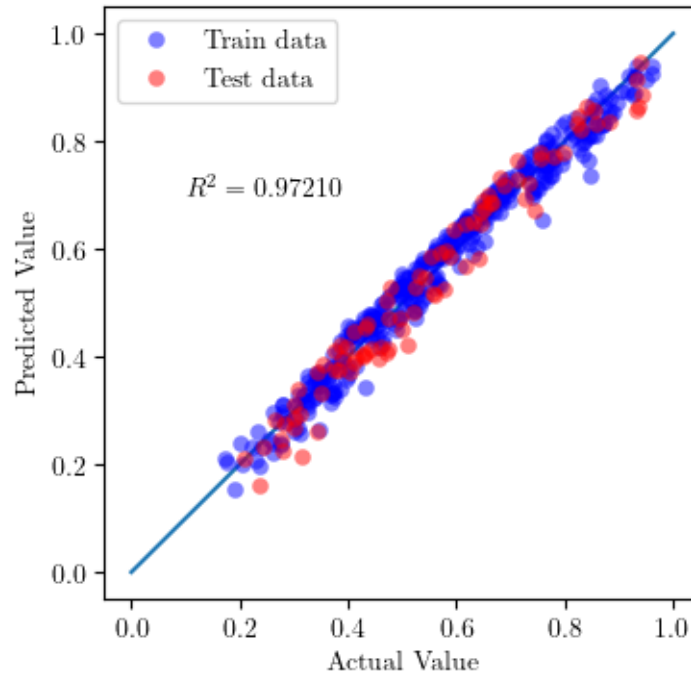
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```
[79]: from sklearn.neural_network import MLPRegressor
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

nn_params = {
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    'mlp__activation' : ['relu']
}

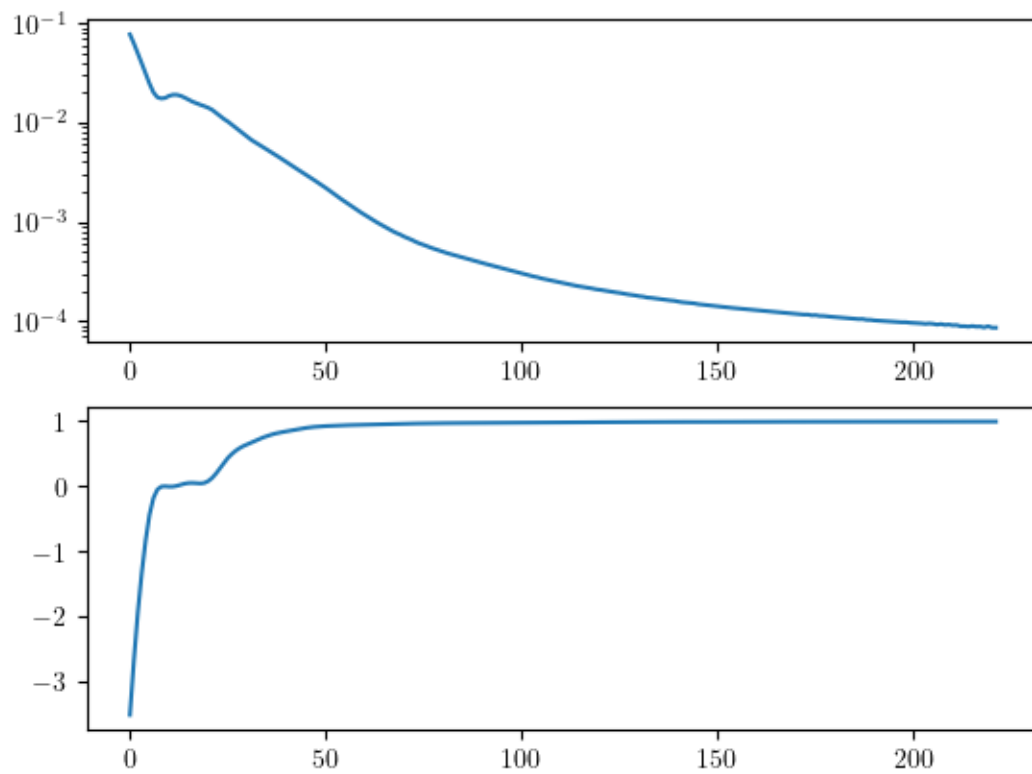
nn_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('mlp', MLPRegressor(random_state=100, early_stopping=True, max_iter=500))
])

nn_cv = GridSearchCV(nn_pipeline, nn_params, scoring='neg_mean_squared_error')
nn_cv.fit(X_train, y_train)

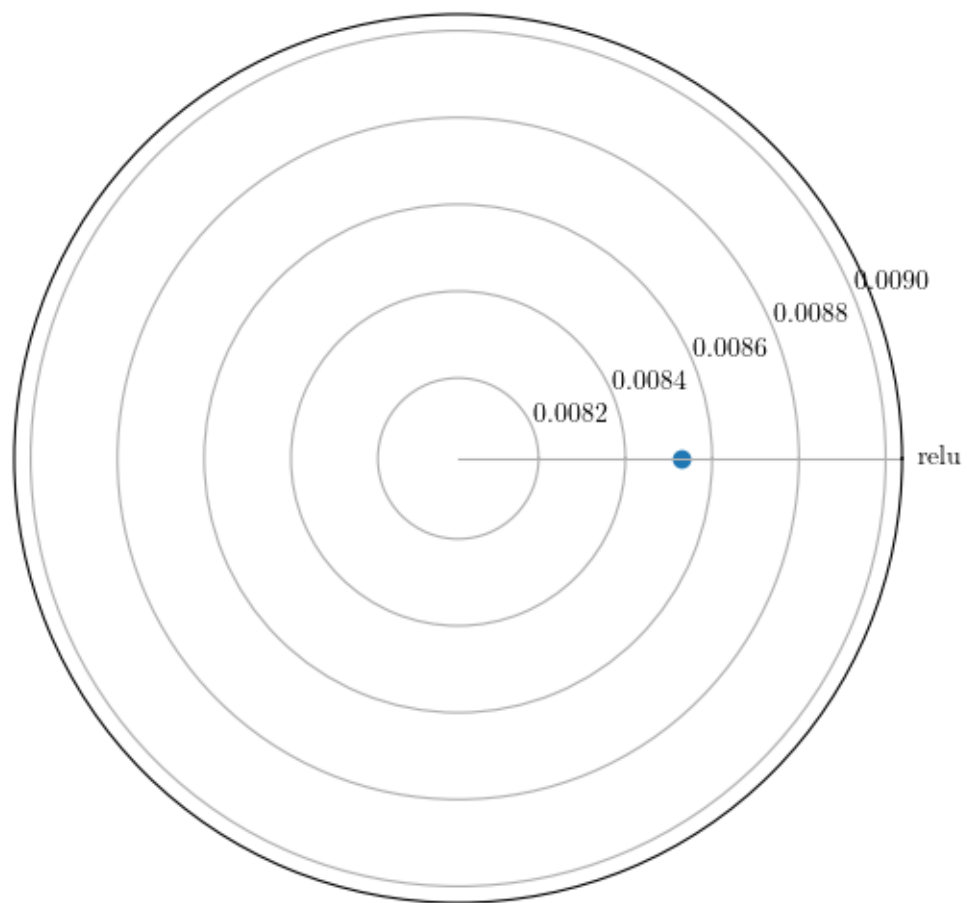
nn_best = nn_cv.best_estimator_
nn_cv_result = pd.DataFrame(nn_cv.cv_results_)

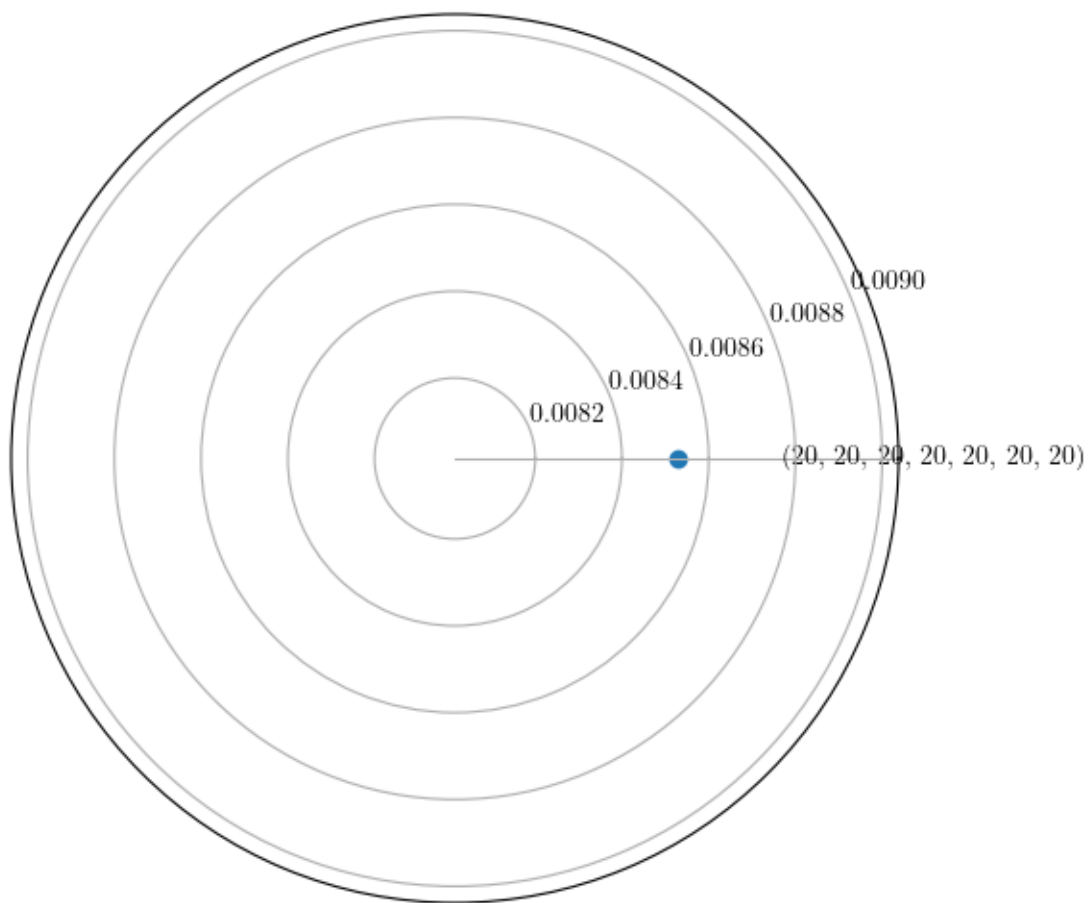
plt.subplot(2, 1, 1)
plt.semilogy(nn_best['mlp'].loss_curve_)
plt.subplot(2, 1, 2)
plt.plot(nn_best['mlp'].validation_scores_)
```

```
plt.savefig('MLP/mlp.png', dpi=640, bbox_inches='tight')
```



```
[80]: plot_cross_validation(nn_cv_result, 'mlp-hyperparams')
```





```
[81]: nn_metrics = evaluate_pipeline(nn_best, 0.3)
      plot_accuracy(nn_metrics, 'MLP')
```

```
MLP
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430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
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287      0.409213
204      0.552481
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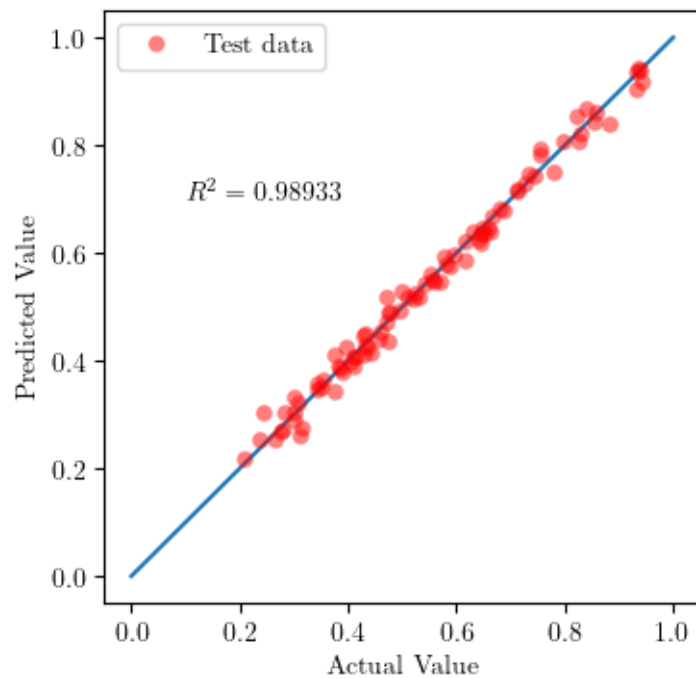
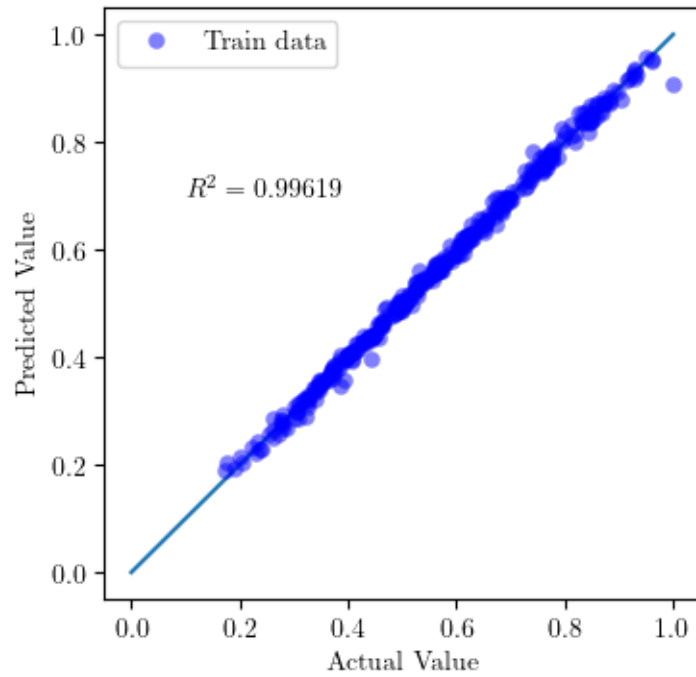
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199 0.352128
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240 0.753010
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328 0.426775

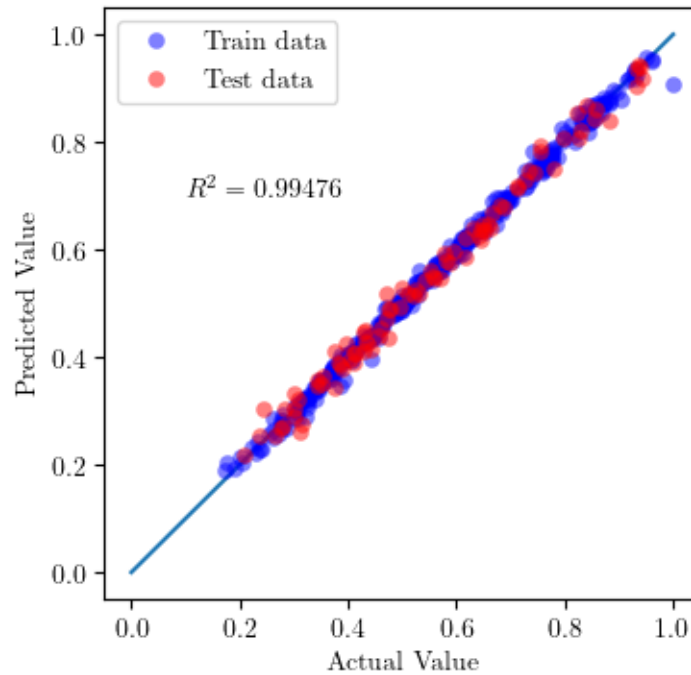
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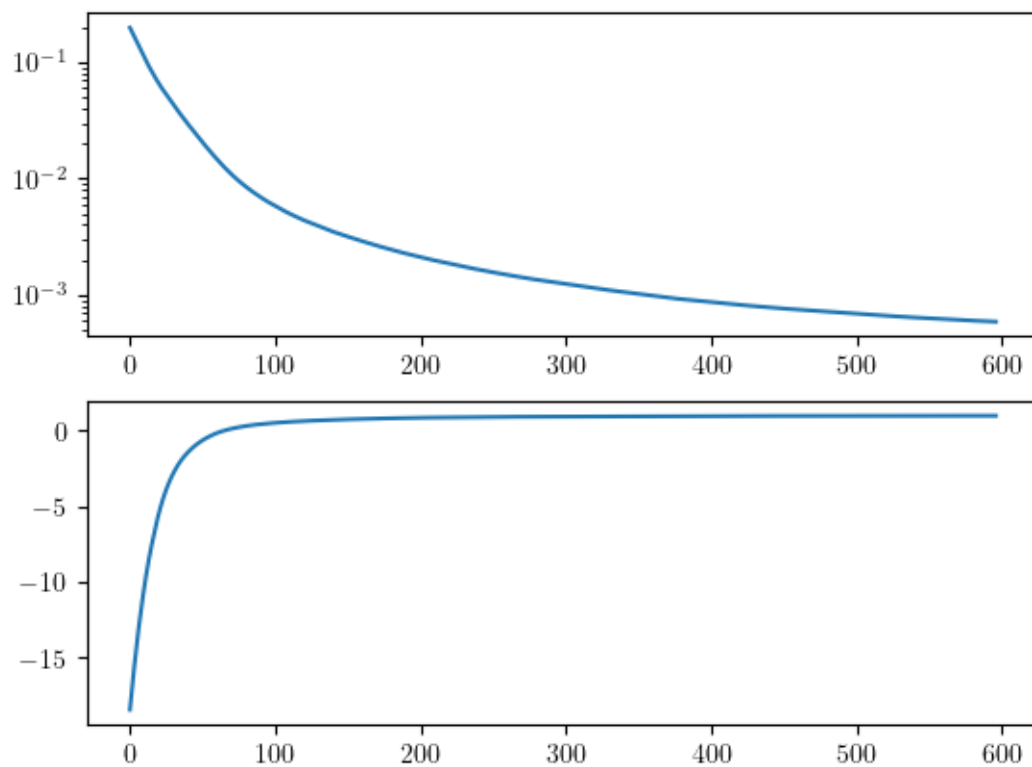
```
[82]: small_nn_params = {
    'mlp_hidden_layer_sizes' : [(x,) for x in range(1, 15)],
    'mlp_activation' : ['relu', 'tanh', 'logistic']
}

small_nn_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('mlp', MLPRegressor(random_state=100, early_stopping=True, max_iter=2000))
])

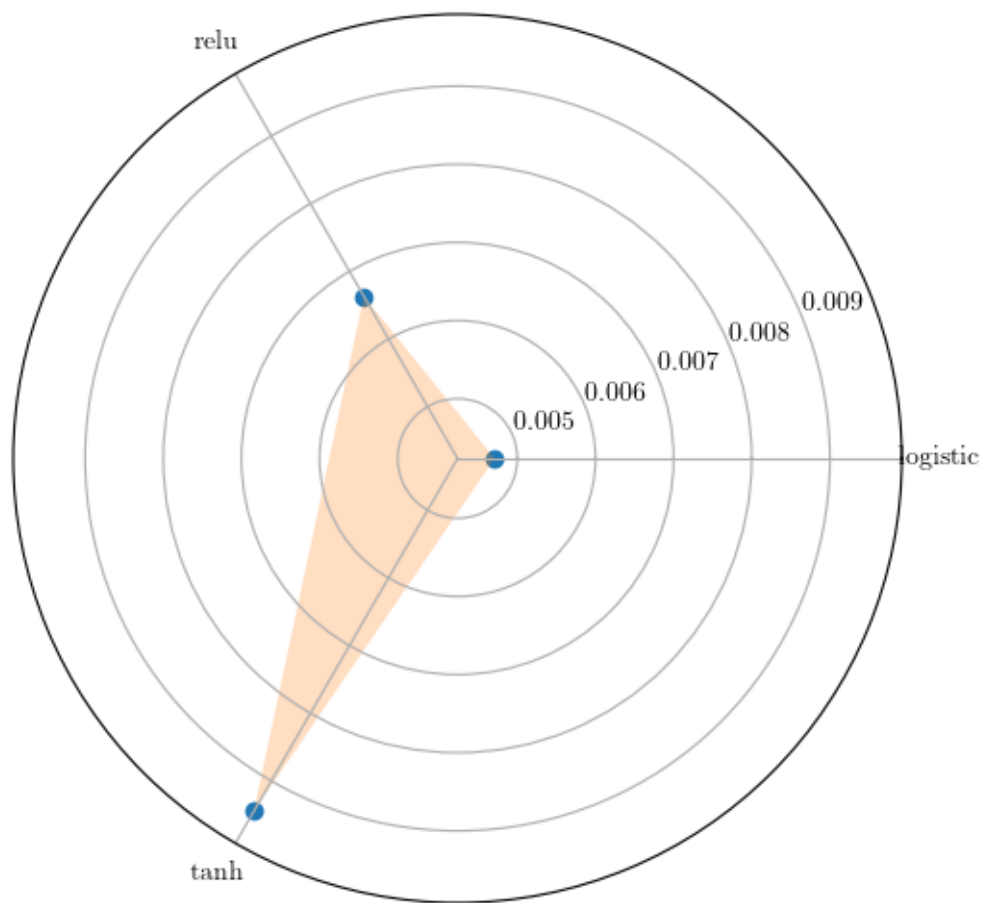
small_nn_cv = GridSearchCV(small_nn_pipeline, small_nn_params,
    scoring='neg_mean_squared_error')
small_nn_cv.fit(X_train, y_train)

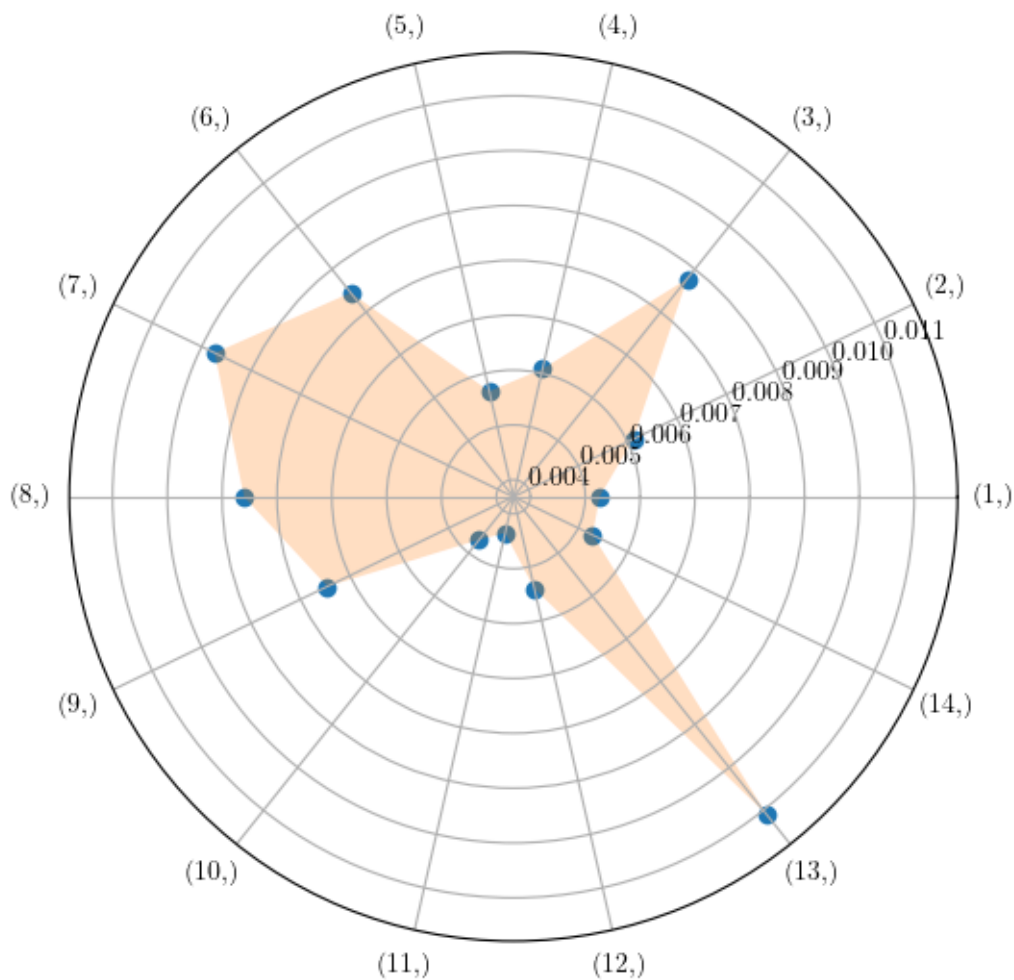
small_nn_best = small_nn_cv.best_estimator_
small_nn_cv_result = pd.DataFrame(small_nn_cv.cv_results_)

plt.subplot(2, 1, 1)
plt.semilogy(small_nn_best['mlp'].loss_curve_)
plt.subplot(2, 1, 2)
plt.plot(small_nn_best['mlp'].validation_scores_)
plt.savefig('small-nn/small-nn.png', dpi=640, bbox_inches='tight')
```



```
[83]: plot_cross_validation(small_nn_cv_result, 'small-nn-hyperparameters')
```





```
[84]: small_nn_metrics = evaluate_pipeline(small_nn_best, 100.0)
      plot_accuracy(small_nn_metrics, 'small-nn')
```

```
small-nn
{'y_train': 21      0.587692
430      0.276756
79       0.630164
408      0.692862
313      0.832889
...
230      0.651196
98       0.689246
322      0.846593
382      0.591085
365      0.558731
Name: Mn/Mp, Length: 352, dtype: float64, 'y_test': 13      0.301036
125      0.643799
```

```

287    0.409213
204    0.552481
172    0.615535

...
199    0.352128
31     0.386317
240    0.753010
59     0.382423
328    0.426775

```

```

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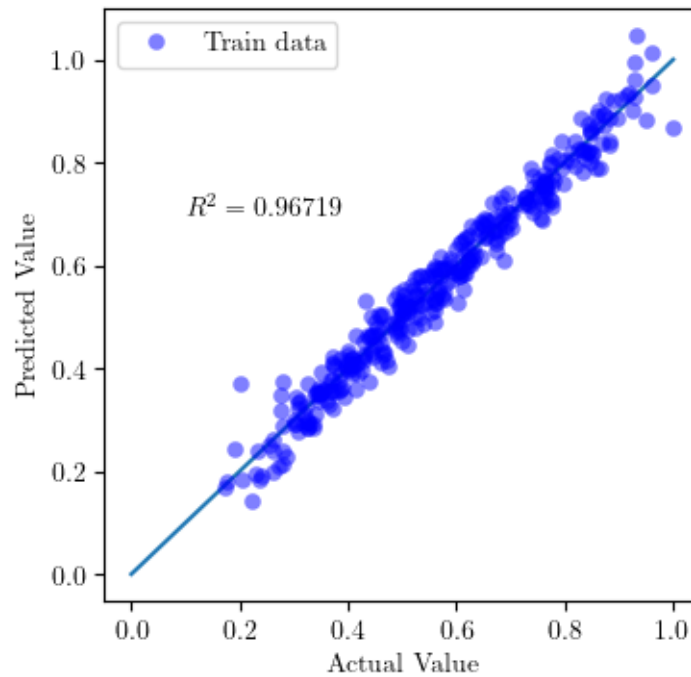
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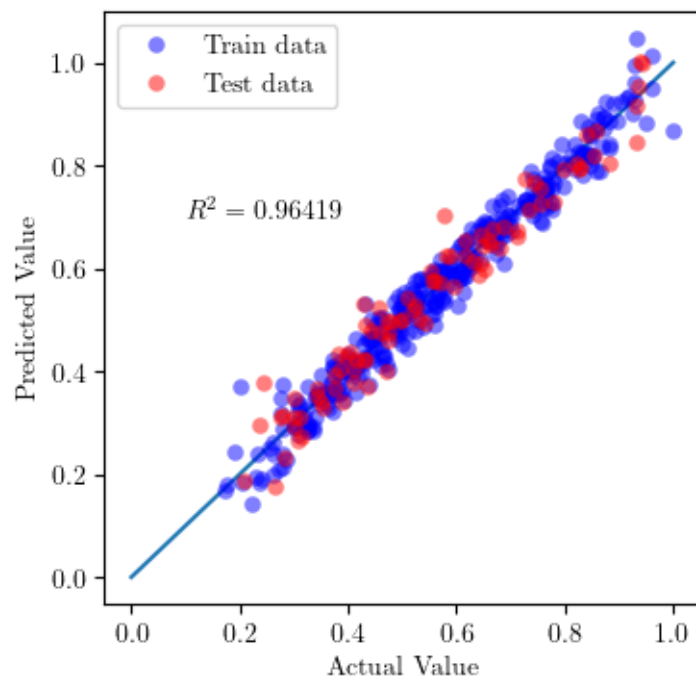
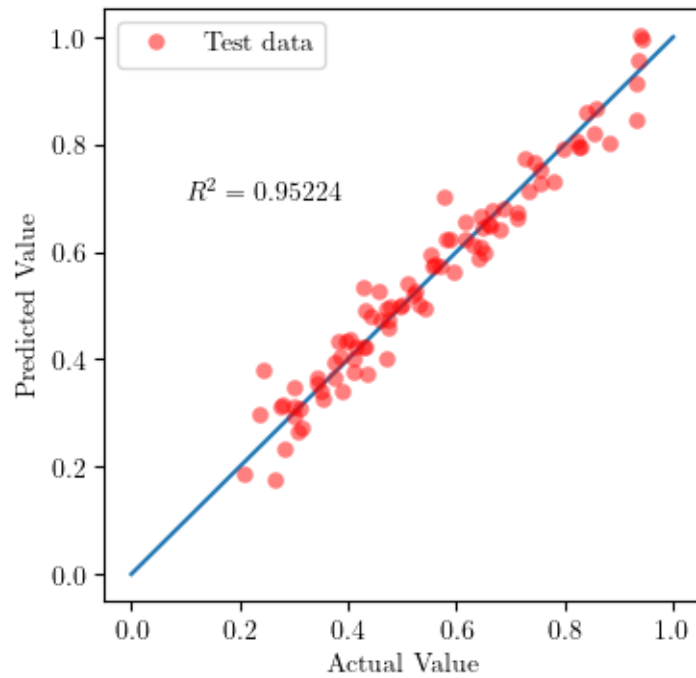
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```
[85]: og_data = pd.read_csv('original.csv')
      og_data.head()
```

```
[85]:
```

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1	139.236	190	15.9	1.77	1.03	42.23	3.49	171595.1703	219030.340139	
2	139.236	230	15.9	1.77	1.03	42.23	3.49	146147.7541	187966.043520	
3	139.236	260	15.9	1.77	1.03	42.23	3.49	128748.2021	155913.110942	
4	139.236	300	15.9	1.77	1.03	42.23	3.49	108993.2842	122477.537470	

	Mn(FE model)	Mp	b	Mcr
0	216638.377325	225562.32	15.9	399132.605082
1	197650.133747	225562.32	15.9	290051.468297
2	170240.744021	225562.32	15.9	205359.253150
3	139104.281134	225562.32	15.9	165618.043830
4	95629.312666	225562.32	15.9	129821.641396

```
[86]: og_data_ratio = pd.DataFrame()
og_data_ratio['Lb/ry'] = og_data['Lb'] / og_data['ry']
og_data_ratio['h/tw'] = og_data['h0'] / og_data['tw']
og_data_ratio['B/2t'] = og_data['b'] / 2 * og_data['tf']
og_data_ratio['h/B'] = og_data['h0'] / og_data['b']
og_data_ratio['tf/tw'] = og_data['tf'] / og_data['tw']
og_data_ratio['E/Fy'] = 29e3 / og_data['Fy (ksi)']
og_data_ratio['Mp/Mcr'] = og_data['Mp'] / og_data['Mcr']
og_data_ratio['Mn/Mp (AISC)'] = og_data['Mn(AISC code)'] / og_data['Mp']
og_data_ratio['Mn/Mp (MOD)'] = og_data['Mn(Modified)'] / og_data['Mp']
```

```
[87]: og_data_ratio.head()
```

```
[87]:
```

	Lb/ry	h/tw	B/2t	h/B	tf/tw	E/Fy	Mp/Mcr	\
0	45.845272	41.0	14.0715	2.655975	1.718447	208.279468	0.565131	
1	54.441261	41.0	14.0715	2.655975	1.718447	208.279468	0.777663	
2	65.902579	41.0	14.0715	2.655975	1.718447	208.279468	1.098379	
3	74.498567	41.0	14.0715	2.655975	1.718447	208.279468	1.361943	
4	85.959885	41.0	14.0715	2.655975	1.718447	208.279468	1.737479	

	Mn/Mp (AISC)	Mn/Mp (MOD)
0	1.000000	0.843663
1	0.971041	0.760744
2	0.833322	0.647926
3	0.691220	0.570788
4	0.542988	0.483207

```
[88]: df.head()
```

```
[88]:
```

	Lb/ry	h/tw	B/2t	h/B	tf/tw	E/Fy	Mp/Mcr	\
0	45.845272	41.0	4.491525	2.655975	1.718447	208.279468	0.558946	
1	54.441261	41.0	4.491525	2.655975	1.718447	208.279468	0.789461	
2	65.902579	41.0	4.491525	2.655975	1.718447	208.279468	0.978898	

3	74.498567	41.0	4.491525	2.655975	1.718447	208.279468	1.248814
4	85.959885	41.0	4.491525	2.655975	1.718447	208.279468	1.607064

	Mn/Mp
0	0.843663
1	0.760744
2	0.647926
3	0.570788
4	0.483207

```
[89]: mse_code = mean_squared_error(og_data_ratio['Mn/Mp (AISC)'], og_data_ratio['Mn/
↳Mp (MOD)'])
r2_code = r2_score(og_data_ratio['Mn/Mp (AISC)'], og_data_ratio['Mn/Mp (MOD)'])
print(mse_code)
print(r2_code)
```

```
0.027484151024631094
0.6181763301765262
```

```
[90]: mae_code = mean_absolute_error(og_data_ratio['Mn/Mp (AISC)'], og_data_ratio['Mn/
↳Mp (MOD)'])
print(mae_code)
```

```
0.12830643132026942
```